

Chapter 6: Electromagnetic Induction

Formula Sheet + Tips & Tricks

Formula Sheet

I. Magnetic Flux and Faraday's Laws

Formula: $\Phi = BA \cos \theta$

Φ = magnetic flux, B = field strength, A = area, θ = angle between B and the normal to the coil.

Formula: $\varepsilon = -N \frac{d\Phi}{dt}$

ε = induced emf (Faraday's Law), N = number of turns, $d\Phi/dt$ = rate of change of flux per turn.

Formula: $q = \frac{N\Delta\Phi}{R}$

q = induced charge, $N\Delta\Phi$ = change in flux linkage, R = total circuit resistance.

Formula: $M = \mu_0 n N A, \quad \varepsilon = M \frac{dI}{dt}$

M = mutual inductance of a small coil (N turns, area A) wound on a long solenoid with n turns/metre.

II. Motional EMF

Formula: $\varepsilon = Bvl$

ε = motional emf, v = velocity of the conductor, l = length of conductor, valid when v , B , l are mutually perpendicular.

Formula: $t = \frac{d}{v}$

Duration for which emf lasts while a loop of side d (measured along the direction of motion) leaves a field region at speed v .

III. Self-Inductance

Formula: $\varepsilon = -L \frac{dI}{dt}$

ε = self-induced emf, L = coefficient of self-induction, dI/dt = rate of change of current.

Formula: $L = \frac{N\Phi}{I}$

L = self-inductance, $N\Phi$ = total flux linkage of the coil, I = current producing it.

Formula: $M = \frac{\mu_0 N_1 N_2 A}{l}$

Mutual inductance of two coils (N_1 , N_2 turns) wound on the same solenoid of length l and cross-section A .

IV. Mutual Induction

Formula: $M = \frac{\phi_1}{I_2}, \quad B_2 = \frac{\mu_0 I_2 R_2^2}{2(R_2^2 + x^2)^{3/2}}$

M = mutual inductance between two coaxial circular loops; R_2 = radius of larger loop, x = distance between centres, found via reciprocity by placing current I_2 in the larger loop and computing flux through the smaller loop.

Note: $\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$ (also written $\text{Wb A}^{-1} \text{ m}^{-1}$ or T m A^{-1} or N A^{-2}).

Tips, Tricks & Hacks

- Lenz's law is just energy conservation in disguise: the induced current always opposes the *change* in flux, never the flux itself — state “opposes the change”, not “opposes the field”, to avoid losing marks.
- $\varepsilon = -\frac{d\phi}{dt}$: the minus sign is Lenz's law baked into the formula; for magnitude-only numericals you can drop it, but never in a conceptual answer.
- Motional emf $\varepsilon = Blv$ applies to a rod moving with velocity $\perp$ to both its length and the field — if the rod moves at an angle, use only the component of velocity $\perp$ to the field and rod.
- Self-inductance $L = \frac{N\phi}{I}$ behaves like “electrical inertia” — it opposes *change* in current, exactly analogous to mass opposing change in velocity.

- Energy stored in an inductor: $U = \frac{1}{2}LI^2$ — structurally identical to $\frac{1}{2}mv^2$ (kinetic energy analogy), useful for quick recall.
- Mutual inductance is symmetric: $M_{12} = M_{21}$, so it doesn't matter which coil you call "primary" when calculating M .
- For a rod sliding on rails (classic EMI problem), always check whether it's asked for emf, current, or the retarding force $F = BIl = \frac{B^2 l^2 v}{R}$ — these look similar but answer different questions.